

Telecom Customer Churn Predictor

An educational walkthrough: data > features > imbalance handling > threshold tuning > prediction.

Data source: [Real Kaggle dataset \(blastchar/telecom-customer-churn\)](#)

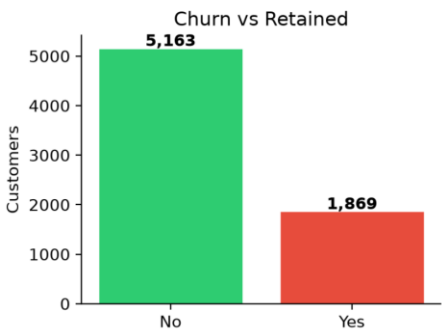
[Data Overview](#) [Feature Engineering](#) [Imbalanced Data](#) [Threshold Tuning](#) [Predict Churn](#)

Raw Data & Churn Distribution

The dataset contains 7,032 customers with 26.6 % churn rate. This imbalance is typical in telecom: most customers stay.

View raw data sample																		
	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecu: 1	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	Payment
0	7596-VHMEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No	No	No	Month-to-month	Yes	Electro
1	5575-QWIDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed
2	3668-QPIBK	Male	0	No	No	2	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-month	Yes	Mailed
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank tr
4	9237-AQGTU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes	Electro
5	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes	Electro
6	1452-KIOKW	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	Month-to-month	Yes	Credit c
7	6713-KOQKC	Female	0	No	No	18	No	No phone service	DSL	Yes	No	No	No	No	No	Month-to-month	No	Mailed
8	7892-POCKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to-month	Yes	Electro
9	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank tr

Churn Distribution



Key Numeric Stats

	tenure	MonthlyCharges	TotalCharges
count	7032	7032	7032
mean	32.42	64.8	2283.3
std	24.55	30.09	2266.77
min	1	18.25	18.8
25%	9	35.59	401.45
50%	29	70.35	1397.48
75%	55	89.86	3794.74
max	72	118.75	8684.8

Churn Rate by Contract Type

